

Scaling Laws of the OMINO-OMICRON-OMNICRON Coordinate Architecture

Abstract

This paper consolidates the scaling laws observed across the OMINO, OMICRON, and OMNICRON model. The architecture begins with Boolean truth-table widths, hexadecimal place-value families, Hamming-style coordinate integrity, factorial permutation rings, Algorithmic Clock fields, Circular Slide Rulers, nested control ranges, and concentric `.omi`, `.imo`, and `.o` artifact surfaces.

The central result is that several apparently unrelated scales are connected by exact arithmetic identities:

$$W = 2^k$$

for the width of a complete Boolean truth-table word with k input variables;

$$2^W = 2^{2^k}$$

for the number of possible complete Boolean functions at that width;

$$(16^n)^p = 2^{4np}$$

for a compound hexadecimal radix;

and therefore:

$$4np = 2^k$$

whenever one compound hexadecimal digit is required to enumerate every complete k -input Boolean function.

The timing geometry introduces a second pair of scales:

$$4320 = 6 \times 3 \times 4 \times 60$$

for the Algorithmic Clock, and:

$$5040 = 7! = 7 \times 6!$$

for the complete Delta rotation ring.

These scales are distinct but coordinated. The Algorithmic Clock answers when a coordinate is active. The Circular Slide Ruler answers which ordering is active. Blackboard projection answers where the coordinate is observed.

The complete coordinate is therefore not a single overloaded number, but a structured triple:

$$X = (p, c, r),$$

where p is projection, c is clock position, and r is rotational order.

1. Introduction

The architecture developed here is based on a simple but powerful principle:

A coordinate system becomes computationally useful when its width, ordering, timing, integrity, and projection laws are all explicit.

A Boolean truth table provides the first scale. For k Boolean input variables, there are:

$$2^k$$

possible input assignments.

A complete Boolean function assigns one output bit to every input assignment. Therefore one complete truth table can be represented as a word of width:

$$W = 2^k.$$

For example:

```
k = 4 → 16-bit truth-table word
k = 5 → 32-bit truth-table word
k = 6 → 64-bit truth-table word
k = 7 → 128-bit truth-table word
k = 8 → 256-bit truth-table word
```

The architecture then adds:

- a circular rolling law over the complete truth-table word;
- hexadecimal factorizations of the word width;
- a Hamming-style coordinate integrity layer;
- a factorial rotation domain;
- a bounded Algorithmic Clock;
- axial and tangential gauge projections;
- nested Metatron and Tetragrammatron fields;
- and three concentric representation surfaces: `.omi`, `.imo`, and `.o`.

These are not independent inventions. They form a coordinated scaling ladder.

2. Truth-Table Scaling

2.1 Assignment count

For k Boolean input variables:

$$W = 2^k$$

is the number of input assignments.

This is also the number of output positions in one complete truth table.

Thus:

$$x \in \{0, 1\}^{2^k}.$$

At $k = 6$:

$$W = 2^6 = 64.$$

Therefore a complete six-input Boolean function is represented by one 64-bit output word.

2.2 Function count

Each of the W output positions can independently contain either zero or one.

Therefore the number of complete Boolean functions is:

$$2^W.$$

Since $W = 2^k$:

$$\boxed{2^{2^k}}$$

is the number of Boolean functions of k variables.

At $k = 6$:

$$2^{2^6} = 2^{64}$$

which is:

$$18,446,744,073,709,551,616.$$

The distinction must remain explicit:

64
positions in one six-input truth-table word

2^{64}
possible complete six-input truth-table words

The machine rolls the positions of one word. It does not physically rotate through all 2^{64} possible words.

3. The General Delta Rolling Law

Let:

$$W = 2^k.$$

A width-dependent Delta transform may be defined as:

$$\Delta_W(x) = \text{rotl}_W(x, 1) \oplus \text{rotl}_W(x, 3) \oplus \text{rotr}_W(x, 2) \oplus C_W.$$

The rotation operators act over the entire W -bit truth-table word.

For $k = 6$:

$$W = 64,$$

so:

$$\Delta_{64}(x) = \text{rotl}_{64}(x, 1) \oplus \text{rotl}_{64}(x, 3) \oplus \text{rotr}_{64}(x, 2) \oplus C_{64}.$$

For $k = 7$:

$$W = 128,$$

so:

$$\Delta_{128}(x) = \text{rotl}_{128}(x, 1) \oplus \text{rotl}_{128}(x, 3) \oplus \text{rotr}_{128}(x, 2) \oplus C_{128}.$$

The same structural rule scales because only the circular width changes.

Two possible state laws must be distinguished:

$$x_{t+1} = \Delta_W(x_t)$$

and:

$$x_{t+1} = x_t \oplus \Delta_W(x_t).$$

The first replaces the state with the transform output. The second applies the transform as a difference mask. They are different dynamical systems and must not be treated as interchangeable.

The width-dependent constant C_W must also have an explicit derivation. A 64-bit constant cannot silently become a 128-bit constant without a declared expansion rule.

4. Four Hexadecimal Scaling Families

4.1 Linear radix family: $16n$

The first family is:

$$B = 16n.$$

Examples:

```
n = 1 → Base 16
n = 2 → Base 32
n = 3 → Base 48
n = 4 → Base 64
```

The number of binary bits required for one radix digit is:

$$w = \lceil \log_2(16n) \rceil.$$

This family is not always perfectly binary aligned.

For example:

```
Base 16 → 4 bits → 16 valid patterns
Base 32 → 5 bits → 32 valid patterns
Base 48 → 6 bits → 48 valid, 16 unused
Base 64 → 6 bits → 64 valid patterns
```

When $16n$ is not a power of two, the architecture needs a policy for the unused bit patterns.

They may be:

- rejected;
- reduced canonically;
- reserved;
- treated as control coordinates;

- or assigned to Omicron or Omnicron gauge roles.

This policy is architectural rather than arithmetic.

4.2 Exponential hexadecimal family: 16^n

The second family is:

$$16^n.$$

Since:

$$16 = 2^4,$$

we have:

$$16^n = 2^{4n}.$$

Therefore one digit in Base- 16^n occupies exactly:

$$4n \text{ bits.}$$

Examples:

$n = 1$	→ Base 16	→ 4 bits
$n = 2$	→ Base 256	→ 8 bits
$n = 4$	→ Base 65,536	→ 16 bits
$n = 8$	→ Base 2^{32}	→ 32 bits
$n = 16$	→ Base 2^{64}	→ 64 bits
$n = 32$	→ Base 2^{128}	→ 128 bits

This family is perfectly aligned with binary storage and is therefore the cleanest fixed-width place-value ladder.

4.3 Sixteenfold compound family: $(16^n)^{16}$

The third family is:

$$(16^n)^{16}.$$

Using exponent laws:

$$(16^n)^{16} = 16^{16n}.$$

Since $16 = 2^4$:

$$16^{16n} = 2^{64n}.$$

Therefore one digit occupies:

$$64n \text{ bits.}$$

Examples:

$n = 1 \rightarrow 64 \text{ bits}$
 $n = 2 \rightarrow 128 \text{ bits}$
 $n = 3 \rightarrow 192 \text{ bits}$
 $n = 4 \rightarrow 256 \text{ bits}$

This family produces a direct ladder of 64-bit register blocks.

It is especially natural when the machine treats each scale as a collection of complete 64-bit truth-table words.

4.4 General compound family: $(16^n)^p$

The most general family is:

$$(16^n)^p.$$

Applying the exponent law:

$$(16^n)^p = 16^{np}.$$

Therefore:

$$(16^n)^p = 2^{4np}.$$

One digit occupies:

$$W = 4np \text{ bits}.$$

Different factorizations can produce the same total width.

For 128 bits:

$$4np = 128,$$

so:

$$np = 32.$$

Possible factorizations include:

$n = 1, p = 32$
 $n = 2, p = 16$
 $n = 4, p = 8$
 $n = 8, p = 4$
 $n = 16, p = 2$
 $n = 32, p = 1$

All describe the same total capacity, but they partition the word into different internal place structures.

Therefore $(16^n)^p$ is not merely a size formula. It is also a factorization language.

5. The Bridge Between Truth Tables and Compound Hexadecimal Bases

A k -input Boolean truth table has width:

$$W = 2^k.$$

A digit in Base- $(16^n)^p$ has width:

$$W = 4np.$$

Equating them gives:

$$4np = 2^k.$$

Therefore:

$$np = 2^{k-2}.$$

Substituting this into the compound radix:

$$(16^n)^p = 2^{4np},$$

we obtain:

$$(16^n)^p = 2^{2^k}.$$

Thus:

$$(16^n)^p = 2^{2^k}$$

whenever:

$$np = 2^{k-2}.$$

This means that one digit in the compound radix has exactly as many possible values as there are Boolean functions of k variables.

The precise interpretation is:

One Base- $(16^n)^p$ digit can enumerate all complete k -input Boolean truth tables when $4np = 2^k$.

Examples:

Inputs k	Truth-table width 2^k	Required np	Function count
4	16 bits	4	2^{16}
5	32 bits	8	2^{32}
6	64 bits	16	2^{64}
7	128 bits	32	2^{128}
8	256 bits	64	2^{256}

This produces three distinct but connected scales:

k
number of Boolean inputs

2^k
width of one truth-table word

$2^{(2^k)}$
number of possible truth-table words

The compound hexadecimal base gives an exact radix representation of the third scale.

6. Omicron's Quaternary Scaling

A separate expression is:

$$\text{Omicron}(k) = (2^2)^k.$$

Therefore:

$$\text{Omicron}(k) = 4^k = 2^{2k}.$$

This is not equal to:

$$2^k$$

or:

$$2^{2^k}.$$

It represents a different combinatorial layer.

At $k = 6$:

$$4^6 = 4096.$$

The three scales are:

binary assignment count:

$$2^6 = 64$$

quaternary Omicron assignment count:

$$4^6 = 4,096$$

complete binary function count:

$$2^{64}$$

A reasonable architectural interpretation is that each of the k positions carries two binary dimensions rather than one.

For example:

bit 1:

value

bit 2:

gauge, relation, orientation, or control state

This would give four states per position:

00

01

10

11

However, the arithmetic alone does not determine the semantics. The role of the second bit must be explicitly declared.

7. Hamming Scaling: Four Controls, Three Relations, Seven Positions

The compact CONS word occupies seven codepoints:

```
0x19 LOGOS
0x1A NOMOS
0x1B FS
0x1C PATHOS
0x1D GS
0x1E RS
0x1F US
```

The architecture divides them into:

```
three relation coordinates:
LOGOS, NOMOS, PATHOS

four scope coordinates:
FS, GS, RS, US
```

The compact word is therefore represented as:

$[7, 4, 3]$.

The parity relations are:

$$s_{\text{LOGOS}} = \text{LOGOS} \oplus \text{FS} \oplus \text{GS} \oplus \text{US},$$

$$s_{\text{NOMOS}} = \text{NOMOS} \oplus \text{FS} \oplus \text{RS} \oplus \text{US},$$

$$s_{\text{PATHOS}} = \text{PATHOS} \oplus \text{GS} \oplus \text{RS} \oplus \text{US}.$$

A valid coordinate satisfies:

$$s_{\text{LOGOS}} = s_{\text{NOMOS}} = s_{\text{PATHOS}} = 0.$$

The seven-position word defines the compact OMICRON Null-Byte Ring. The eight-position COBS-CONS completion defines the OMNICRON Gauge Ring. The six-position operational projection defines OMINO.

The nested ladder is:

```
OMNICRON [8,4,4]
  completed gauge ring

OMICRON [7,4,3]
  compact null-byte-relative ring

OMINO [6,4,2]
  operational tangential selector
```

The `[6,4,2]` label is architectural rather than a claim that OMINO is a conventional named Hamming code.

8. Factorial Scaling and Rotation

8.1 Four-scope permutation kernel

The four scope coordinates:

```
FS
GS
RS
US
```

can be ordered in:

$$4! = 24$$

distinct ways.

Thus:

$$\boxed{24}$$

is the local scope-order resolution.

This explains the decomposition:

$$32 = 24 + 8.$$

The 24-position body represents the complete ordering environment of the four scope dimensions. The 8-position terminal represents the completed COBS-CONS ring.

8.2 Full seven-position rotation

The complete compact word has seven positions.

All complete orderings form the symmetric group S_7 :

$$|S_7| = 7! = 5040.$$

Therefore the Circular Slide Ruler coordinate is:

$$r \in \{0, \dots, 5039\}.$$

Each value of r identifies one ordering of:

LOGOS
NOMOS
FS
PATHOS
GS
RS
US

The codepoints remain immutable. The rotation coordinate changes only their circular order.

8.3 Anchored reduction

The identity:

$$7! = 7 \times 6!$$

gives an anchored interpretation.

Choose one of the seven positions as the current anchor. The remaining six positions can be ordered in:

$$6! = 720$$

ways.

Therefore:

$$7 \times 720 = 5040.$$

The complete seven-position ring exceeds the anchored six-position operational field by exactly:

$$720 = 6!.$$

This gives the relation:

OMICRON

complete seven-position ruler

OMINO

six-position operational field after one anchor is fixed

9. Algorithmic Clock Scaling

The Algorithmic Clock is not the 5040-position rotation ring.

It is constructed from:

6 operational OMINO orientations

3 relation intonations

4 Blackboard quadrants

60 sexagesimal positions

Therefore:

$$4320 = 6 \times 3 \times 4 \times 60.$$

The clock coordinate is:

$$c \in \{0, \dots, 4319\}.$$

It may be represented as:

$$c = (o, \lambda, q, t),$$

where:

$$o \in \{0, \dots, 5\},$$

$$\lambda \in \{\text{LOGOS}, \text{NOMOS}, \text{PATHOS}\},$$

$$q \in \{0, 1, 2, 3\},$$

and:

$$t \in \{0, \dots, 59\}.$$

The flattened coordinate is:

$$c = ((o \cdot 3 + \lambda) \cdot 4 + q) \cdot 60 + t.$$

This produces:

$$0 \leq c < 4320.$$

The inverse decoding is:

$$t = c \bmod 60,$$

$$c_1 = \left\lfloor \frac{c}{60} \right\rfloor,$$

$$q = c_1 \bmod 4,$$

$$c_2 = \left\lfloor \frac{c_1}{4} \right\rfloor,$$

$$\lambda = c_2 \bmod 3,$$

$$o = \left\lfloor \frac{c_2}{3} \right\rfloor.$$

10. The Relationship Between 4320 and 5040

The two totals can be written:

$$5040 = 7 \times 3 \times 4 \times 60,$$

$$4320 = 6 \times 3 \times 4 \times 60.$$

Their shared factors are:

3	LOGOS, NOMOS, PATHOS
4	quadrants or control dimensions
60	sexagesimal positions

Their difference is the distinction between seven complete positions and six anchored operational positions.

The arithmetic difference is:

$$5040 - 4320 = 720.$$

Since:

$$720 = 6!,$$

the complete ruler contains one additional anchored six-position permutation domain beyond the operational clock field.

The two coordinates answer different questions:

- r
Which ordering is selected?
- c
At which relational clock position is that ordering active?

The same rotation may occur at different clock positions:

$$r_A = r_B, \quad c_A / c_B. \quad =$$

The same clock position may expose different rotations:

$$c_A = c_B, \quad r_A / r_B. \quad =$$

Therefore rotation and synchrony must not be collapsed.

11. Projection Scaling

The architecture adds a third coordinate:

$$p.$$

This is the Blackboard projection selector.

The complete state is:

$$X = (p, c, r).$$

The three components answer:

- p
where the coordinate is viewed
- c
when the coordinate is active

r
which ordering is active

This is one of the most important scaling rules because it prevents an address, clock, and permutation index from being overloaded into one number.

The coordinate model is therefore not merely temporal or spatial. It is projective, synchronous, and rotational.

12. Gauge Scaling

The operational OMINO gauge uses six low-plane prefixes:

2
3
4
5
6
7

and six high-plane prefixes:

A
B
C
D
E
F

The low axial origins are:

0x20
0x30
0x40
0x50
0x60
0x70

The low tangential completions are:

0x2F
0x3F
0x4F
0x5F
0x6F
0x7F

The high axial origins are:

0xA0
0xB0
0xC0
0xD0
0xE0
0xF0

The high tangential completions are:

0xAF
0xBF
0xCF
0xDF
0xEF
0xFF

Each row has the form:

X0 ... XF

where:

X0
zero-relative axial origin

XF
complete four-place tangential closure

The suffix **F** corresponds to:

1111₂,

the complete four-place Metatron field.

13. Nested Metatron and Tetragrammatron Scaling

The terminal coordinate lens is nested:

```
0x18...0x1F
  eight-position COBS-CONS field

0x19...0x1F
  seven-position CONS word

0x1B...0x1F
  five-position Tetragrammatron block

0x1C...0x1F
  four-position Metatron field
```

The nested widths are:

8, 7, 5, 4.

They correspond to different validation jurisdictions.

Metatron

Metatron occupies:

0x1C ... 0x1F.

Its four place anchors are:

```
0x0001
0x0010
0x0100
0x1000
```

Together they produce:

```
1111
```

or:

$0xF$.

Metatron therefore witnesses four-place incidence and position closure.

Tetragrammatron

Tetragrammatron occupies:

$0x1B \dots 0x1F$.

Its witness anchors include:

```
0x1E
0x3C
0x78
0x00
```

It governs the relation between the inner incidence positions and the return to the zero polynomial.

The nesting law is:

Metatron validates the four-place incidence field contained inside the five-position Tetragrammatron relation field.

14. Gnomonic Azimuth Scaling

The Gnomonic Projection Azimuth is represented by:

$0xAA55$.

It supports two readings:

```
nibble scale:
A | A | 5 | 5

byte scale:
AA | 55
```

The nibble reading corresponds to four-place Metatron resolution.

The byte reading corresponds to two-part Tetragrammatron resolution.

The relation:

$$A \oplus 5 = F$$

connects complementary alternating patterns to complete four-bit closure.

Thus $0x4A55$ is an orientation bridge between incidence-scale and relation-scale readings.

15. Truth-Table, Control, and Gauge Scaling

The six operational OMINO coordinates produce:

$$2^6 = 64$$

binary assignment positions.

The four persistent scope dimensions produce:

$$2^4 = 16$$

control quotients.

All functions over the six-coordinate truth-table surface form:

$$2^{64}$$

possible complete functions.

The scales are:

```

4 control coordinates
  16 control assignments

6 operational coordinates
  64 truth-table rows

64 output positions
  2^64 complete truth functions

```

This creates an internal ladder:

$$\begin{aligned}
 4 &\longrightarrow 16, \\
 6 &\longrightarrow 64, \\
 64 &\longrightarrow 2^{64}.
 \end{aligned}$$

The first arrow is a binary assignment expansion. The second is another binary assignment expansion. The third counts all output labelings over the resulting table.

16. Artifact Scaling

The architecture contains five semantic families:

```
RULES
FACTS
CLOSURES
COMBINATORS
CONS
```

Each appears across three concentric surfaces:

```
.omi
  user declaration

.imo
  system definition

.o
  machine executable coordinate
```

Therefore:

$$5 \times 3 = 15.$$

This produces a fifteen-position artifact environment.

The three surfaces are not necessarily a compilation chain. They are three complete descriptions of the same semantic family at different depths.

The concentric direction is:

```
outside:
.omi

middle:
.imo

inside:
```

.0

centroid:
0x00 / 0°

The scaling law is therefore:

$$\text{semantic families} \times \text{representation depths} = 5 \times 3 = 15.$$

17. Local Incidence Scaling

The geometry uses:

4320,

4320²,

and:

4320⁴.

These may be read as:

4320
local incidence field

4320²
paired or surface incidence field

4320⁴
complete four-domain consistency horizon

The exact values are:

$$4320^2 = 18,662,400,$$

and:

$$4320^4 = 348,285,173,760,000.$$

These scales increase the incidence domain without changing the local coordinate law.

The transform remains relative to the fixed OMINO centroid:

$$(0x00, 0^\circ).$$

The scaling is therefore geometric rather than procedural: larger incidence surfaces reuse the same branchless coordinate law.

18. Unified Scaling Table

Domain	Law	Meaning
Boolean assignments	2^k	truth-table word width
Boolean functions	2^{2^k}	all complete truth tables
Quaternary Omicron	4^k	four-state assignment layer
Hex place	$16^n = 2^{4n}$	$4n$ -bit radix digit
Compound hex place	$(16^n)^p = 2^{4np}$	$4np$ -bit radix digit
Truth-table bridge	$4np = 2^k$	compound width equals truth-table width
Scope permutations	$4! = 24$	FS/GS/RS/US orderings
Anchored operations	$6! = 720$	six-position permutation domain
Full rotations	$7! = 5040$	complete seven-position orderings
Algorithmic Clock	$6 \times 3 \times 4 \times 60 = 4320$	bounded synchrony field
Artifact environment	$5 \times 3 = 15$	five terms across three surfaces
Control quotients	$2^4 = 16$	FS/GS/RS/US assignments
OMINO rows	$2^6 = 64$	six-selector truth-table positions
Local incidence	4320	base clock/incidence field
Surface incidence	4320^2	paired incidence surface
Horizon incidence	4320^4	four-domain consistency horizon

19. Canonical Rules

The observed scaling rules can be stated canonically.

Rule 1: Truth-table width

$$W = 2^k$$

A k -input Boolean function is represented by a 2^k -bit truth-table word.

Rule 2: Function-space cardinality

$$F_k = 2^{2^k}$$

There are 2^{2^k} Boolean functions of k inputs.

Rule 3: Rolling domain

$$\Delta_W : \{0, 1\}^W \rightarrow \{0, 1\}^W$$

The Delta law rolls one complete truth-table word of width W .

Rule 4: Hexadecimal digit width

$$16^n = 2^{4n}$$

A Base- 16^n digit occupies $4n$ bits.

Rule 5: Compound hexadecimal width

$$(16^n)^p = 2^{4np}$$

A Base- $(16^n)^p$ digit occupies $4np$ bits.

Rule 6: Truth-table/radix equivalence

$$4np = 2^k$$

When this holds, one compound hexadecimal digit has exactly enough states to enumerate every complete k -input Boolean function.

Rule 7: Scope-order resolution

$$4! = 24$$

The four FS/GS/RS/US coordinates define 24 local orderings.

Rule 8: Complete rotation

$$7! = 5040$$

The seven-position compact word defines 5040 complete orderings.

Rule 9: Anchored operational reduction

$$7! = 7 \times 6!$$

Fixing one anchor exposes a six-position domain of 720 orderings.

Rule 10: Algorithmic Clock

$$4320 = 6 \times 3 \times 4 \times 60$$

The clock combines OMINO orientation, relation intonation, quadrant, and sexagesimal tick.

Rule 11: Complete coordinate

$$X = (p, c, r)$$

Projection, synchrony, and rotation are independent coordinates.

Rule 12: Integrity before admission

A coordinate is admitted only after its Hamming, Metatron, Tetragrammatron, and Azimuth witnesses satisfy the active profile.

Rule 13: Concentric representation

$$5 \times 3 = 15$$

Five semantic families across `.omi`, `.imo`, and `.o` form the complete fifteen-position artifact environment.

20. Implementation Consequences

A practical implementation can use these laws in stages.

Stage 1: truth-table codec

Implement:

$$\text{width}(k) = 2^k$$

and pack one Boolean function as a 2^k -bit word.

Stage 2: Delta transform

Implement width-generic circular rotations:

```
rotr_W  
rotr_W
```

and the selected Delta transition law.

Stage 3: factorial ranking

Implement ranking and unranking for:

```
S4 → 24 scope orders  
S6 → 720 anchored operations  
S7 → 5040 complete rotations
```

Stage 4: clock codec

Implement packing and unpacking of:

```
(o, λ, q, t)  
↔  
c ∈ [0, 4319]
```

Stage 5: Hamming integrity

Implement:

```
LOGOS  
NOMOS  
PATHOS  
FS  
GS  
RS  
US
```

with syndrome generation and correction.

Stage 6: gauge selection

Implement the low and high axial/tangential rows.

Stage 7: nested witnesses

Implement Metatron, Tetragrammatron, and Azimuth validation.

Stage 8: compound radix profiles

Implement the parameterized width:

$$W = 4np$$

and require:

$$W = 2^k$$

when the compound hexadecimal place is intended to enumerate one complete truth-function space.

Stage 9: artifact binding

Bind the same coordinate semantics across:

```
.omi  
.imo  
.o
```

without allowing one representation surface to redefine the others.

21. Status of the Claims

The observed claims should be classified carefully.

Arithmetic identities

The following are exact:

$$2^k,$$

$$\begin{aligned}
&2^{2^k}, \\
&16^n = 2^{4n}, \\
&(16^n)^p = 2^{4np}, \\
&4! = 24, \\
&6! = 720, \\
&7! = 5040, \\
&4320 = 6 \times 3 \times 4 \times 60, \\
&5040 - 4320 = 720.
\end{aligned}$$

Defined architectural laws

The following are model definitions:

```

OMNICRON [8,4,4]
OMICRON [7,4,3]
OMINO [6,4,2]
X = (p,c,r)
Metatron field = 0x1C...0x1F
Tetragrammatron block = 0x1B...0x1F
Azimuth = 0xAA55

```

Implementation choices still requiring a canonical profile

The following must be frozen explicitly:

```

the width-dependent constant C_W

whether:
x' = Delta_W(x)
or:
x' = x XOR Delta_W(x)

the exact bit packing of p, c, r, Hamming data, and gauge data

the canonical factorization (n,p) when multiple factorizations
produce the same width

the exact runtime acceptance masks for Metatron,
Tetragrammatron, and Azimuth

```

the treatment of unused patterns in Base-(16n)

the role of the second bit in the quaternary 4^k Omicron layer

22. Conclusion

The architecture exhibits a consistent hierarchy of scaling laws.

At the Boolean level:

$$k \rightarrow 2^k \rightarrow 2^{2^k}.$$

At the hexadecimal level:

$$(n, p) \rightarrow 4np \rightarrow 2^{4np}.$$

The two ladders meet when:

$$4np = 2^k.$$

At the control level:

$$4 \rightarrow 4! = 24.$$

At the operational level:

$$6 \rightarrow 6! = 720.$$

At the complete rotation level:

$$7 \rightarrow 7! = 5040.$$

At the clock level:

$$6 \times 3 \times 4 \times 60 = 4320.$$

At the representation level:

$$5 \times 3 = 15.$$

At the coordinate level:

$$X = (p, c, r).$$

These are not all measurements of the same thing.

They describe different structural dimensions:

2^k
truth-table width

$2^{(2^k)}$
function-space cardinality

$4np$
hexadecimal place width

24
local scope-order resolution

720
anchored operational permutation domain

4320
bounded Algorithmic Clock

5040
complete Circular Slide Ruler

15
concentric artifact environment

p
projection

c
synchrony

r
rotation

The principal law of the architecture is therefore not that every scale collapses into one number.

It is that every scale has a distinct role, and exact arithmetic identities allow those roles to be coordinated without confusing them.

The resulting model can support a deterministic coordinate codec in which:

1. Boolean inputs determine the truth-table word width.
2. The Delta law rolls the complete word.
3. Hexadecimal parameters factor that width into place-value layers.
4. Hamming coordinates establish integrity.

5. Metatron establishes incidence closure.
6. Tetragrammatron establishes relation closure.
7. Azimuth establishes orientation.
8. The Algorithmic Clock establishes synchrony.
9. The Circular Slide Ruler establishes ordering.
10. Blackboard projection establishes the observation surface.
11. `.omi`, `.imo`, and `.o` expose the same coordinate at user, system, and executable depths.

The unified scaling statement is:

$$W = 2^k = 4np, F_k = 2^W = (16^n)^p, c \in \{0, \dots, 4319\}, r \in \{0, \dots, 5039\}, X = (p, c, r).$$

This gives the architecture a precise mathematical spine: truth-table width, function-space cardinality, hexadecimal factorization, clock synchrony, rotational order, and projective placement remain separate, but they are connected by explicit scaling laws.